

BUSINESS ANALYTICS

MINI PROJECT

On

ZOMATO SALES ANALYSIS



Submitted by:

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Abstract:

The Zomato app analysis dataset provides insights into restaurant attributes, customer preferences, and service quality. This study examines 50 restaurant listings, covering parameters such as location, ratings, food categories, pricing, discounts, delivery charges, and service quality. The key findings highlight the impact of factors like rating trends, pricing strategies, and delivery efficiency on customer satisfaction. The dataset also captures aspects like dine-in capacity, payment modes, and special offerings such as happy hours. By leveraging Business Analytics techniques, this analysis aims to identify patterns that can optimize restaurant performance, enhance customer experience, and inform strategic decision-making within the food delivery ecosystem.

Introduction:

The food delivery industry has witnessed rapid growth, with platforms like Zomato revolutionizing customer dining experiences. Understanding key factors influencing restaurant performance, customer preferences, and service quality is essential for both business owners and data analysts. This study utilizes a dataset of 50 restaurant listings from the Zomato app to explore critical aspects such as restaurant ratings, menu diversity, pricing strategies, delivery services, and customer engagement. By analysing these parameters, we aim to uncover patterns that contribute to customer satisfaction, operational efficiency, and competitive advantage in the online food delivery market. This analysis provides valuable insights for restaurant owners, business analysts, and decision-makers looking to optimize service offerings and enhance customer experiences.

Project Objectives

- Data Preparation & Loading: Importing and cleaning sales-related data.
- Building Data Relationships: Connecting multiple datasets for meaningful insights.
- DAX Calculations & Measures: Implementing key performance metrics such as total items sold, average sales.
- Dashboard Creation & Visualization: Designing an interactive dashboard using various chart types, slicers, and KPIs to enhance data interpretation.
- Custom Themes & UI Improvements: Improving dashboard aesthetics by applying custom colors, fonts, and formatting.

This project serves as a practical implementation of Power BI's capabilities in the financial domain, demonstrating how data-driven decisions can improve sales processes.

Data Collection:

The dataset used for this analysis was collected from the Zomato app, focusing on various attributes of restaurants listed on the platform. The data was gathered based on the following criteria:

1. Restaurant Information:

- Name and location of the restaurant
- Type of food offered (Veg/Non-Veg)
- Best-rated food items

2. Customer Experience Metrics:

- Ratings and reviews
- Delivery service quality
- Happy hours and discounts

3. Operational Details:

- Number of items available
- Starting price of food items
- Delivery charges and dine-in capacity
- Payment modes (Cash, Online, etc.)

4. Special Features:

- Pets-friendly status
- Takeaway and dine-in availability

The data was manually extracted from Zomato and Google by observing restaurant listings in different locations. Some values, such as ratings and pricing, were dynamically updated, ensuring the dataset reflects real-world trends in customer preferences and business performance. This dataset forms the foundation for analysing business insights and improving restaurant operations in the competitive food delivery industry.

Data Processing:

The collected metrics such as restaurant, location, and engagement rates and other features are cleaned and analyzed to identify significant trends.

Importing Data into Power BI

1. Open Power BI and import the .csv file containing Zomato data.
2. Load the dataset by selecting the required sheet.
3. Navigate to the **Table View** section to inspect the data structure.

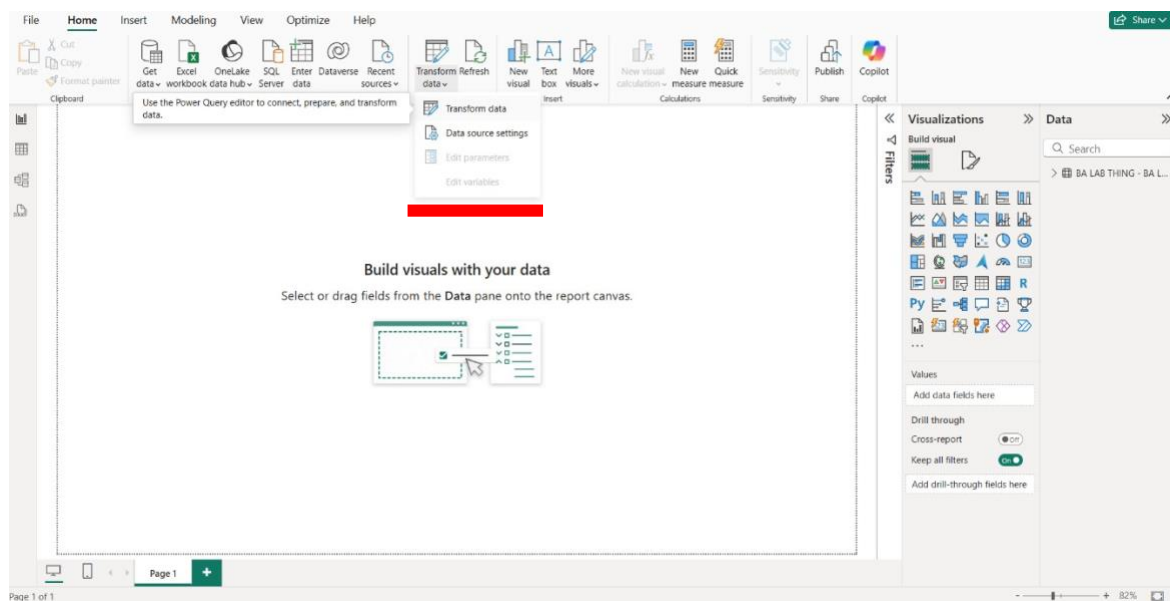
BA LAB THING - BA LAB THING.csv

File Origin: 1252: Western European (Windows) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

restaurant	locations	rating	veg	non-veg	best rated	items	Starting price	discount	delivery charges	timing
1	Cp	4	Y	Y	Chicken Biryani	100	150	10.00%	50- 150	9:00 - 1
2	Anekal	5	N	Y	Ghee Podi rice	456	70	10.00%	50- 151	9:00 - 1
3	EC -PH1	3.4	Y	Y	Poha	435	60	10.00%	50- 152	9:00 - 1
4	EC-PH2	5	N	Y	Upma	234	50	10.00%	50- 153	9:00 - 1
5	bommasandra	4.5	Y	N	Paneer Butter Masala	432	200	12.00%	50- 154	9:00 - 1
6	bommanhalli	3.9	Y	N	Palak Paneer	123	123	13.00%	50- 155	9:00 - 1
7	silk board	4	N	Y	Chicken kebab	234	150	13.00%	50- 156	9:00 - 1
8	BTM	3	Y	Y	Champaran mutton	321	300	13.00%	50- 157	9:00 - 1
9	Madiwala	2	Y	Y	Shushi	100	400	13.00%	50- 158	9:00 - 1
10	kormangla	5	N	Y	Mach bhaat	532	300	13.00%	50- 159	9:00 - 1
11	Jaynagar	2	N	Y	Chicken curry	543	250	13.00%	50- 160	9:00 - 1
12	JP nagar	4	N	Y	Chicken Tandoori	432	187	15.00%	50- 161	9:00 - 1
13	Shanti Nagar	3.2	Y	N	Litti Chokha <3	23	50	15.00%	50- 162	9:00 - 1
14	TC palya	3	Y	N	Rasmalayi	23	30	15.00%	50- 163	9:00 - 1
15	KR puram	4	Y	N	Ghewar	41	100	15.00%	50- 164	9:00 - 1
16	Tin factor	5	N	Y	Prons fry	123	225	15.00%	50- 165	9:00 - 1
17	White Field	2	Y	Y	Pomphret	234	222	15.00%	50- 166	9:00 - 1
18	Marathalli	1	Y	Y	Woodfire Pizza	213	100	15.00%	50- 167	9:00 - 1
19	Kalyannagar	3	N	Y	Momos	45	60	15.00%	50- 168	9:00 - 1
20	Attibelle	4	N	Y	Noodles	324	50	15.00%	50- 169	9:00 - 1

Extract Table Using Examples | **Load** | Transform Data | Cancel

- Go to the **Transform data** section.
- Navigate to the **Data** part and transform it.



In the query editor:

- **Delete unwanted columns** that are not required for analysis.
- **Rename necessary columns** to maintain clarity and uniformity in the dataset.

- Once all modifications are made, select **Copy** to save a backup if needed.
- Finally, click **Apply** to save and implement the changes in the table.

The screenshot displays the Microsoft Power BI Desktop interface. The main view shows a data table with the following columns: restaurant, locations, rating, veg, non-veg, best rated, and items. The table contains 27 rows of data. The right sidebar shows the 'Query Settings' pane with 'APPLIED STEPS' including 'Source', 'Promoted Headers', and 'Changed Type'.

restaurant	locations	rating	veg	non-veg	best rated	items
1	1 Cp	4	Y	Y	Chicken Biryani	
2	2 Anekal	5	N	Y	Ghee Podi rice	
3	3 EC-PH1	3.4	Y	Y	Poha	
4	4 EC-PH2	5	N	Y	Uppma	
5	5 bommasandra	4.5	Y	N	Paneer Butter Masala	
6	6 bommanhalli	3.9	Y	N	Palak Paneer	
7	7 silk board	4	N	Y	Chicken kebab	
8	8 BTM	3	Y	Y	Champan mutton	
9	9 Madiwala	2	Y	Y	Shushi	
10	10 kormangla	5	N	Y	Mach bhaat	
11	11 Jaynagar	2	N	Y	Chicken curry	
12	12 JP nagar	4	N	Y	Chicken Tandoori	
13	13 Shanti Nagar	3.2	Y	N	Litti Chokha <3	
14	14 TC palya	3	Y	N	Rasmalai	
15	15 KR puram	4	Y	N	Ghewar	
16	16 Tin factor	5	N	Y	Prons fry	
17	17 White Field	2	Y	Y	Pomphret	
18	18 Marathalli	1	Y	Y	Woodfire Pizza	
19	19 Kalyannagar	3	N	Y	Momos	
20	20 Attibelle	4	N	Y	Noodles	
21	21 Chruv Street	5	Y	N	Paneer Tikka	
22	22 Vijay Nagar	3	N	Y	Chicken noodles	
23	23 Sanjay.N	2	Y	Y	Gatte ki Sabji	
24	24 Yellahanka	2	N	Y	Chicken Dumplings	
25	25 Kasturi Nagar	3	Y	Y	Soya bean chillie	
26	26 Malleswaram	4	N	Y	Mutton keema	
27	27 Baswagudi	5	Y	N	Gobhi Manchurian	

- The data has been cleaned and organized to ensure it's accurate and consistent. Duplicate values have been removed, unwanted columns have been deleted, and the important columns have been renamed for clarity. Now, the data is properly structured and ready for analysis. It's prepared for creating useful charts, graphs, and reports that will offer valuable insights and help in making better decisions.

Tables, Columns and Measures

Objective: Creating calculated tables, columns, and measures to enhance data analysis.

Steps:

1. Creating Calculated Columns:

- Add a new column for total Sales value: = SUMPRODUCT(G2:G51, H2:H51)
- Standard deviation of total number of items =
STDEV.S(G2:G51)/SQRT(COUNT(G2:G51))
- Sample Error of total number of items = H2 - AVERAGE(H2:H51)

- Sample Variance =VAR.S(R2:R51)

POSSIBLE DATA LOSS Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format.

C52 fx =STDEV.S(C2:C51)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
9	28	Hoskutte	3	N	Y	Lobsters	34	76	5%	50-177	9:00-12:0	2hr	best	6	all types	y	n	150	y				
0	29	Majestic	2.5	Y	Y	Gulab jami	43	45	5%	50-178	9:00-12:0	2hr	best	7	all types	y	n	160	n				
1	30	HRBR	3	N	Y	Shrimps	43	222	5%	50-179	9:00-12:0	2hr	best	8	all types	n	y	240	n				
2	31	HSR	4	N	Y	Goan Style	433	333	5%	50-180	9:00-12:0	2hr	best	9	all types	n	y	255	y				
3	32	Sarjapur	5	N	Y	Fish Curry	222	123	5%	50-181	9:00-12:1	2hr	best	2	all types	n	y	321	y				
4	33	chikpette	3	N	Y	Chicken cu	234	321	5%	50-182	9:00-12:1	2hr	best	3	all types	n	y	442	n				
5	34	Mg road	4.9	N	Y	Roghan jos	321	231	7%	50-183	9:00-12:1	2hr	best	5	all types	n	y	111	n				
6	35	Bannergha	5	Y	N	Kaju Katli	32	111	7%	50-184	9:00-12:1	2hr	best	2	all types	n	y	189	n				
7	36	Jigani	3.3	Y	N	Chhat	32	123	7%	50-185	9:00-12:1	1hr	best	2	all types	n	y	89	y				
8	37	Iggalur	3.7	Y	Y	Masala Ch	321	123	7%	50-186	9:00-12:1	1hr	best	5	all types	n	y	76	y				
9	38	Suryacity	2	Y	N	Aalu tikka	123	145	7%	50-187	8:00-02:0	1hr	best	4	all types	n	y	87	n				
0	39	VBHC	1.9	Y	N	Mushroom	321	179	7%	50-188	8:00-02:0	1hr	best	6	all types	y	y	55	y				
1	40	Hosur	4	Y	N	Ghee Idli	12	200	7%	50-189	8:00-02:0	1hr	average	7	all types	y	y	45	n				
2	41	Kundanah	3	N	Y	Mutton bir	34	150	7%	50-190	8:00-02:0	1hr	average	2	all types	y	y	320	y				
3	42	Hebbali	2	N	Y	Chicken Kc	21	260	14%	50-191	8:00-02:0	1hr	average	3	all types	y	y	229	y				
4	43	hebgodi	1	Y	Y	Khurma	12	250	14%	50-192	8:00-02:0	1hr	average	4	all types	y	y	299	y				
5	44	Mullur	2	Y	Y	Rajma Cha	123	45	14%	50-193	8:00-02:0	1hr	average	2	all types	y	y	199	y				
6	45	Indira Nag	3	Y	Y	Rassam Ri	123	50	14%	50-194	8:00-02:0	1hr	average	5	all types	y	n	189	n				
7	46	Gandhi Na	4	N	Y	Bisibelle Bi	90	10	14%	50-195	8:00-02:0	1hr	average	6	all types	y	n	155	n				
8	47	Akshay Na	2	Y	N	Lasgna	80	20	14%	50-196	8:00-02:0	1hr	average	2	all types	y	n	432	n				
9	48	Mahadevi	4.3	N	Y	Burger	80	30	14%	50-197	8:00-02:1	1hr	average	4	all types	n	y	212	y				
0	49	Trinity	4.2	Y	N	Ragi mudd	78	40	14%	50-198	8:00-02:1	1hr	average	5	all types	n	y	321	y				
1	50	Kudlur gate	5	N	Y	Momos	78	300	14%	50-199	8:00-02:1	1hr	average	6	all types	n	y	432	n				
2			1.155176																				
3																							
4																							

BA_MINI_PROJECT

2. DAX Measures:

- Average Rating =AVERAGE(BA_MINI_PROJECT[rating])
- Average Discount =
AVERAGEX(BA_MINI_PROJECT,VALUE(SUBSTITUTE(BA_MINI_PROJECT[discount], "%", "")))
- Total Items Sold = SUM(BA_MINI_PROJECT[items])
- Total Revenue = SUMX(BA_MINI_PROJECT, BA_MINI_PROJECT[items] * BA_MINI_PROJECT[Starting price])

Table: BA_MINI_PROJECT (50 rows) Column: Average Rating (0 distinct values)

restaurant	locations	rating	veg	non-veg	best rated	items	Starting price	discount	delivery charges	timing	happy hrs	Delivery service	portions	pay
1	Cp	4	Y	Y	Chicken Biryani	100	150	0.1	50-150	9:00-10:00	3hr	excellent	3	Cr
2	Anekal	5	N	Y	Ghee Podi rice	456	70	0.1	50-151	9:00-10:01	3hr	excellent	4	Cr
3	EC-PH1	3.4	Y	Y	Poha	435	60	0.1	50-152	9:00-10:02	3hr	excellent	6	al
4	EC-PH2	5	N	Y	Upma	234	50	0.1	50-153	9:00-10:03	3hr	excellent	8	al
5	bommanasandra	4.5	Y	N	Paneer Butter Masala	432	200	0.12	50-154	9:00-10:04	3hr	excellent	4	al
6	bommanahalli	3.9	Y	N	Palak Paneer	123	123	0.13	50-155	9:00-10:05	3hr	excellent	3	al
7	silk board	4	N	Y	Chicken kebab	234	150	0.13	50-156	9:00-10:06	3hr	excellent	4	al
8	BTM	3	Y	Y	Champan mutton	321	300	0.13	50-157	9:00-10:07	3hr	excellent	4	al
9	Madiwala	2	Y	Y	Shushi	100	400	0.13	50-158	9:00-10:08	3hr	excellent	3	al
10	kormangla	5	N	Y	Mach bhaat	532	300	0.13	50-159	9:00-10:09	3hr	excellent	2	al
11	Jaynagar	2	N	Y	Chicken curry	543	250	0.13	50-160	9:00-10:10	3hr	excellent	2	al
12	JP nagar	4	N	Y	Chicken Tandoori	432	187	0.15	50-161	9:00-10:11	3hr	excellent	3	al
13	Shanti Nagar	3.2	Y	N	Litti Chokha <3	23	50	0.15	50-162	9:00-10:12	3hr	excellent	4	al
14	TC palaya	3	Y	N	Rasmalai	23	30	0.15	50-163	9:00-10:13	3hr	excellent	5	al
15	KR puram	4	Y	N	Ghevar	41	100	0.15	50-164	9:00-10:14	3hr	excellent	6	al
16	Tin factor	5	N	Y	Prons fry	123	225	0.15	50-165	9:00-10:15	3hr	good	7	al
17	White Field	2	Y	Y	Pomphret	234	222	0.15	50-166	9:00-10:16	3hr	good	8	al
18	Marathalli	1	Y	Y	Woodfire Pizza	213	100	0.15	50-167	9:00-10:17	3hr	good	1	al
19	Kalyannagar	3	N	Y	Momos	45	60	0.15	50-168	9:00-10:18	3hr	good	2	al
20	Attibelle	4	N	Y	Noodles	324	50	0.15	50-169	9:00-10:19	3hr	good	3	al
21	Churuch Street	5	Y	N	Paneer Tikka	100	88	0.15	50-170	10:00-10:00	3hr	good	4	al
22	Vijay Nagar	3	N	Y	Chicken noodles	201	90	0.15	50-171	9:00-12:00	3hr	good	5	al
23	Sanjay N	2	Y	Y	Gatte ki Sabji	234	86	0.05	50-172	9:00-12:01	3hr	good	7	al
24	Yellahanka	2	N	Y	Chicken Dumplings	123	189	0.05	50-173	9:00-12:02	2hr	good	2	al
25	Kasturi Nagar	3	Y	Y	Soya bean chillie	335	77	0.05	50-174	9:00-12:03	2hr	good	3	al
26	Malleswaram	4	N	Y	Mutton keema	435	92	0.05	50-175	9:00-12:04	2hr	best	4	al

Building a Complete Dashboard in Power BI Using Different Visualization Techniques

Objective:

Create an interactive dashboard with multiple visualizations.

Steps:

Key Metrics (Cards):

Total sales

Sum of Capacity

Average rating

- **Bar Chart:** Sum of portions by Sum of Capacity.
- **Pie Chart:** Sum of Restaurant by Sum of Locations
- **Line Chart:** Average of Delivery Charges • **Table Visualization:** Detailed Sales

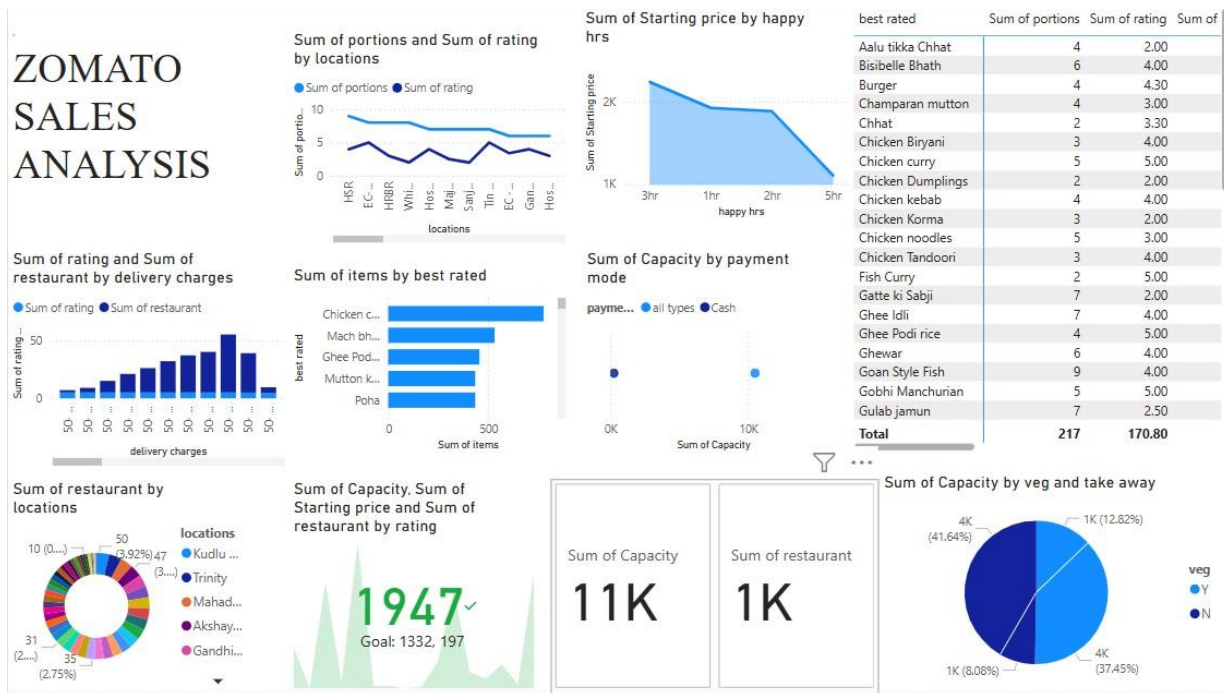
Analysis

- **Filters & Slicers:**

○ Sales Status ○

Location ○ Customer

Best Rated



Tools Used:

- **Microsoft Excel:** Used for organizing, cleaning, and analyzing raw data to derive insights.
- **Business Intelligence (BI) Tools:** Platforms such as Tableau or Power BI help create interactive dashboards for a clearer understanding of advertising performance.

Conclusion:

The Zomato data analysis using Power BI has provided valuable insights into various aspects of the restaurant and food delivery business. Through interactive visualizations and key metrics, we have identified significant trends in customer preferences, order patterns, and restaurant performance across different locations.

Key findings from the analysis include:

- **Customer Demographics:** Insights into popular cuisine types based on customer age, location, and spending behavior.
- **Restaurant Performance:** Identification of high-performing restaurants based on ratings, reviews, and order volumes, along with areas where improvements can be made.
- **Order Trends:** A detailed breakdown of order frequency, peak hours, and seasonal fluctuations, enabling better resource allocation and inventory planning.
- **Customer Satisfaction:** Analysis of review ratings and comments, helping pinpoint factors that influence customer satisfaction.

By leveraging Power BI's interactive dashboards, the data can be continuously updated to track performance, make data-driven decisions, and refine strategies to enhance customer experience and business profitability.